

2.1 (a)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

$$\bar{x} = 1$$

$$\bar{y} = 2$$

$$(b) b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8, \quad b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 = 1.2$$

b_1 是指当 $x=0$, fitted line 之截距为 1.2.

b_2 是 fitted line 之斜率为 0.8.

$$(c) \sum_{i=1}^5 x_i^2 = 15, \quad \sum_{i=1}^5 x_i y_i = 18.$$

$$\sum_{i=1}^5 x_i^2 - N \bar{x}^2 = 15 - 5 \times 1^2 = 10 = \sum_{i=1}^5 (x_i - \bar{x})^2$$

$$\sum_{i=1}^5 x_i y_i - N \bar{x} \bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum_{i=1}^5 (x_i - \bar{x})(y_i - \bar{y})$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$s_y^2 = \sum_{i=1}^5 \frac{(y_i - \bar{y})^2}{N-1} = \frac{4+1+1+4}{5-1} = \frac{10}{4} = 2.5$$

$$s_x^2 = \sum_{i=1}^5 \frac{(x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

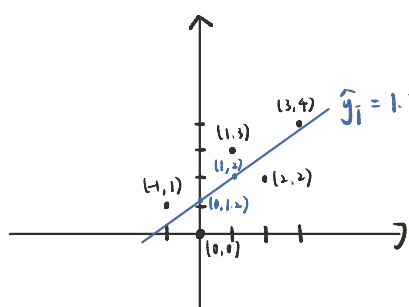
$$s_{xy} = \sum_{i=1}^5 \frac{(y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = s_{xy} / s_x s_y = 2 / 2.5 = \frac{4}{5} = 0.8$$

$$CV_x = 100(s_x / \bar{x}) = 100\sqrt{2.5} = 158.11388.$$

$$\text{median}(x) = 1$$

(e)



$$\hat{y}_i = 1.2 + 0.8x_i$$

$$\hat{y}_i = 1.2 + 0.8x_i$$

$$x=0 \Rightarrow \hat{y}_i = 1.2$$

$$x=1 \Rightarrow \hat{y}_i = 2$$

(f) 由 (e) 可知, $(\bar{x}, \bar{y}) = (1, 2)$ 在 fitted line 上.

(g) $\bar{x} = 1$ 代入 fitted line $\Rightarrow 1.2 + 0.8 = 2 = \bar{y} \Rightarrow$ on the fitted line.

$$(h) \bar{\hat{y}} = \sum \hat{y}_i / N = 10/5 = 2 = \bar{y}$$

$$(i) \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$$

$$(j) \hat{\sigma}_{b_2|X} = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12, \quad \text{se}(b_2) = \sqrt{\hat{\sigma}_{b_2|X}} = \sqrt{0.12} = 0.3464$$

(a) Call:
lm(formula = totalscore ~ small, data = star_a)

Residuals:

	Min	1Q	Median	3Q	Max
	-189.44	-50.44	-9.44	43.06	324.38

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.675	249.396	<2e-16 ***
small	12.175	5.369	2.268	0.0236 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared: 0.006608, Adjusted R-squared: 0.005323
F-statistic: 5.142 on 1 and 773 DF, p-value: 0.02363

$\hat{\beta}_1 = 916.442$: The students' combined aptitude scores is 916.442 in the regular-sized class.

$\hat{\beta}_2 = 12.175$: Students in small classes scored, on average, 12.175 points higher than those in regular-sized classes. This effect is statistically significant.

In conclusion, the small class can improve learning performance effectively.

(b) Call:
lm(formula = readscore ~ small, data = star_a)

Residuals:

	Min	1Q	Median	3Q	Max
	-74.665	-21.590	-2.665	16.410	194.335

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.488	290.760	< 2e-16 ***
small	6.924	2.174	3.185	0.00151 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared: 0.01295, Adjusted R-squared: 0.01167
F-statistic: 10.14 on 1 and 773 DF, p-value: 0.001507

$$\hat{\beta}_1 = 432.665 \quad \hat{\beta}_2 = 6.924$$

According to the READSCORE results, the average total score is 432.665 in regular-sized class, and small class score 6.924 points higher, which is statistically significant.

In conclusion, small-class teaching has a statistically significant positive impact on reading scores compared to math scores.

Call:
lm(formula = mathscore ~ small, data = star_a)

Residuals:

	Min	1Q	Median	3Q	Max
	-144.777	-35.028	-5.028	29.223	142.223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.449	197.545	<2e-16 ***
small	5.251	3.578	1.467	0.143

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared: 0.002778, Adjusted R-squared: 0.001488
F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427

$$\hat{\beta}_1 = 483.777 \quad \hat{\beta}_2 = 5.251$$

According to the MATHSCORE results, the average total score is 483.777 in regular-sized class, and small class score 5.251 points higher, which is not statistically significant.

(c) Call:
lm(formula = totalscore ~ aide, data = star_c)

Residuals:

	Min	1Q	Median	3Q	Max
	-191.748	-50.442	-5.442	40.558	312.558

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.559	257.529	<2e-16 ***
aide	4.306	4.994	0.862	0.389

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared: 0.0008898, Adjusted R-squared: -0.0003068
F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888

$$\hat{\beta}_1 = 916.442, \quad \hat{\beta}_2 = 4.306 \text{ (not significant)}$$

We found that regular-sized classes with a teacher aide scored 4.306 points higher than those without an aide. However, this difference is not statistically significant.

The small class has better learning performance than the teacher aide.

(d) Call:
lm(formula = readscore ~ aide, data = star_c)

Residuals:

	Min	1Q	Median	3Q	Max
	-74.665	-21.536	-2.536	15.464	194.335

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.460	296.341	<2e-16 ***
aide	2.871	2.049	1.401	0.161

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared: 0.002347, Adjusted R-squared: 0.001152
F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615

$$\hat{\beta}_1 = 432.665, \quad \hat{\beta}_2 = 2.871 \text{ (not significant)}$$

While the presence of a teacher aide is associated with slightly higher scores in both reading (+2.871) and math (+1.435), neither effect is statistically significant. This suggests that adding a teacher aide to a regular-sized class is less effective than reducing the class size.

In conclusion, the small class with a better learning performance than the teacher aide whether the subject or sized in class.

Call:
lm(formula = mathscore ~ aide, data = star_c)

Residuals:

	Min	1Q	Median	3Q	Max
	-146.212	-31.212	-5.777	29.223	142.223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.355	205.464	<2e-16 ***
aide	1.435	3.304	0.434	0.664

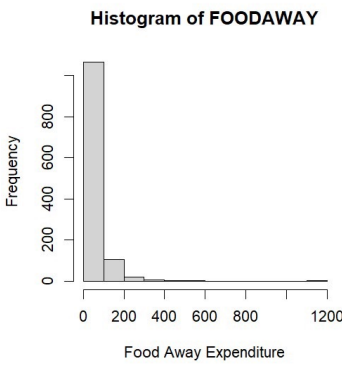
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared: 0.0002258, Adjusted R-squared: -0.0009715
F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642

$$\hat{\beta}_1 = 483.777, \quad \hat{\beta}_2 = 1.435 \text{ (not significant)}$$

2.25

(a)



```
> summary(cex5_small$foodaway)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.00  12.04   32.55   49.27   67.50  1179.00
```

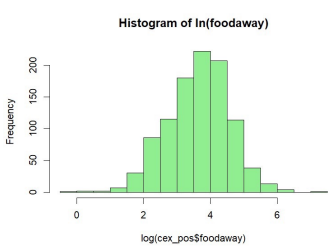
Mean is 49.27 and median is 32.55.
25th and 75th percentiles are 12.04 and 67.5.

(b)

```
> aggregate(foodaway ~ college, data = cex5_small, mean)
  college foodaway
1      0 49.56999
2      1 48.59718
> aggregate(foodaway ~ college, data = cex5_small, median)
  college foodaway
1      0  28.89
2      1  36.11
> aggregate(foodaway ~ advanced, data = cex5_small, mean)
 advanced foodaway
1      0 42.76161
2      1 73.15494
> aggregate(foodaway ~ advanced, data = cex5_small, median)
 advanced foodaway
1      0  28.89
2      1  48.15
> no_degree_data <- subset(cex5_small, advanced == 0 & college == 0)
> summary(no_degree_data$foodaway)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.00   9.63   26.02   39.01   52.65   437.78
```

	median	mean
college	36.11	48.597
advanced degree	48.15	73.15
no advanced or college	26.02	39.01

(c)



```
summary(log(cex5_small$foodaway))
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -Inf  2.488   3.483   -Inf  4.212   7.072
```

foodaway : 1200 observations
ln(foodaway) : 1022 observations

The number of observations for ln(foodaway) is fewer than foodaway because the log-transformation excludes any observations with a value of zero, as ln(0) is undefined.

(d)

```
Call:
lm(formula = log(foodaway) ~ income, data = cex_pos)

Residuals:
    Min       1Q   Median       3Q      Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

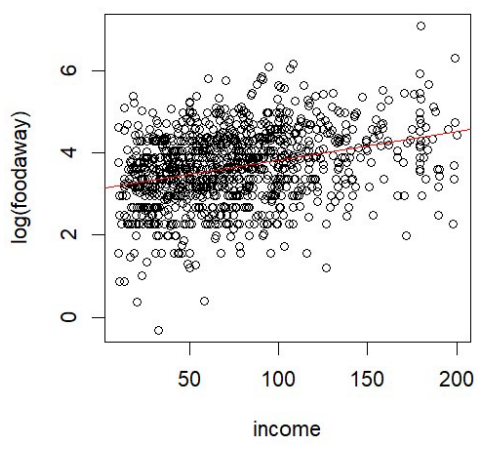
Coefficients:
(Intercept) 3.1293004 0.0565503 55.34 <2e-16 ***
income      0.0069017 0.0006546 10.54 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826, Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16
```

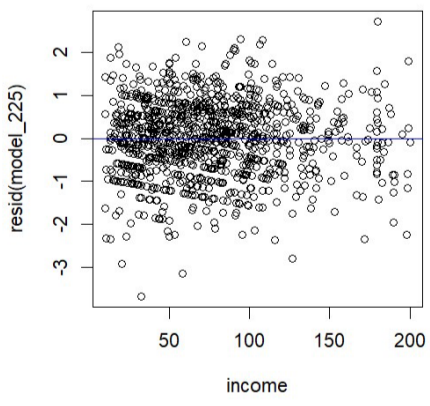
slope : 0.0069017

When income increases by 1 unit (\$100), the food expenditure increases 0.69% on average.

(e)



(f)



Because the residual plot shows that the residuals are randomly scattered around the zero baseline without any discernible pattern or systematic trend, it indicates that the residuals are randomly distributed and the model is appropriately specified.

```

# 2.22 code
setwd("G:/我的雲端硬碟/交大/碩一下/econometric/PoE5data")
load("star5_small.rdata")

# (a)
star_a <- subset(star5_small, aide == 0)
model_a <- lm(totalscore ~ small, data = star_a)
summary(model_a)

# (b) read vs math
summary(lm(readscore ~ small, data = star_a))
summary(lm(mathscore ~ small, data = star_a))

# (c) 助教
star_c <- subset(star5_small, small == 0)
model_c <- lm(totalscore ~ aide, data = star_c)
summary(model_c)

# (d)
summary(lm(readscore ~ aide, data = star_c))
summary(lm(mathscore ~ aide, data = star_c))

# 2.25 code
setwd("G:/我的雲端硬碟/交大/碩一下/econometric/PoE5data")
load("cex5_small.rdata")

# (a)
hist(cex5_small$foodaway, main="Histogram of FOODAWAY", xlab="Food Away Expenditure")
summary(cex5_small$foodaway)

# (b)
aggregate(foodaway ~ college, data = cex5_small, mean)
aggregate(foodaway ~ college, data = cex5_small, median)

aggregate(foodaway ~ advanced, data = cex5_small, mean)
aggregate(foodaway ~ advanced, data = cex5_small, median)

no_degree_data <- subset(cex5_small, advanced == 0 & college == 0)
summary(no_degree_data$foodaway)

# (c)
hist(log(cex5_small$foodaway), main="Histogram of ln(foodaway)", col="lightgreen")
summary(log(cex5_small$foodaway))
sum(cex5_small$foodaway == 0) # 被剔除掉的人數 (即等於 0 的人數)
sum(cex5_small$foodaway > 0) # 取對數後「剩下」的有效樣本數 (即大於 0 的人數)

# (d)
cex_pos <- subset(cex5_small, foodaway > 0)
model_225 <- lm(log(foodaway) ~ income, data = cex_pos)
summary(model_225)

# (e)
plot(log(foodaway) ~ income, data = cex_pos)
abline(model_225, col="red")

#(f)
plot(resid(model_225) ~ income, data = cex_pos)
abline(h=0, col="blue")

```